Cloud Gate

Retrieval Pipeline Review
PalmX POC (Proof of Concept)
V 1.0

1. Document Details

1.1 Document Information

Document Title	Retrieval Pipeline Review PalmX Proof of Concept (POC)
Document Owner	CloudGate LLC
Document Classification	Palmx Confidential

1.2 Document Change Record

Date	Author	Title	Version	Change Reference
30/03/2025	Pragadeswaran MS	Senior AI Consultant	1.0	Initial Release

1.3 Document Reviewers

Date	Reviewer	Title	Version
30/03/2025	Amr Ali	Final Version	1.0

2. Contents

1. DOCUMENT DETAILS 2

1.1 Document Information 2

1.2 Document Change Record 2

1.3 Document Reviewers 2

3. INTRODUCTION TO CLOUDGATE 4

4. OBJECTIVE 5

5. DETAILED FINDINGS & RESPONSES 5

5.1 Reranker Model & Effectiveness 5

5.2 TOP-K RETRIEVAL BEHAVIOR 5
5.2.1 Key Observations: *6*
5.3 CHUNKING STRATEGY 6
5.3.1 Implications: *6*
5.4 CONTEXT VS METADATA 6
5.4.1 Observation: *7*
5.5 RETRIEVAL APPROACH & DUPLICATES 7
5.5.1 Structured Filters *7*
5.5.2 Query Representation *8*
5.5.3 Limitation: *8*
5.6 ROOT CAUSE ANALYSIS 8
5.6.1 Key contributing factors: *8*
6. SUMMARY 8

3. Introduction to CloudGate

CloudGate delivers digital transformation and technology services from ideation to execution, enabling Global clients to outperform the competition. CloudGate is "Born digital," and takes an agile, collaborative approach to creating customized solutions across the digital value chain. At the same time, our deep expertise in infrastructure and applications management helps optimize your IT into a strategic asset.

Whether you need to differentiate your company, reinvent business functions, or accelerate revenue growth, we can get you there. Digital transformation is powered by Cloud Transformation and CloudGate delivers a holistic approach to cloud transformation-from advisory to build, and from migration to management-that accelerates a company's move to digital business

Services Portfolio:

- **Training**
- **Managed Services**
- **Licensing**
- **Consultancy**
- **Professional Services**
- **Security Assessment**
- **Modern Work**
- **Application Modernization**
- **Data & AI**
- **Business Applications**
- **Security**
- **Cloud Migration**

4. Objective

To evaluate the current retrieval pipeline, including query handling, filtering, and ranking mechanisms, in order to identify gaps in relevance, accuracy, and system performance, and to highlight key areas for improvement.

5. Detailed Findings & Responses

5.1 Reranker Model & Effectiveness

Based on the current implementation, our team confirms that there is **no dedicated machine learning–based reranker** (such as a cross-encoder or external reranking API) integrated into the system.

At present, the pipeline operates using:

- **Dense vector retrieval (FAISS)** for initial candidate generation
- **Fuzzy string matching (RapidFuzz)** on project IDs as a secondary matching mechanism

It is important to note that this fuzzy matching step does **not function as a true semantic reranker**. Additionally:

- There is **no score fusion or reordering mechanism** combining multiple relevance signals
- There is **no evaluation framework** (e.g., NDCG, MRR, A/B testing) currently implemented to measure improvement over baseline retrieval

5.2 Top-K Retrieval Behavior

The system currently follows this retrieval flow:

- **Initial retrieval:** Top *2K* candidates (e.g., 6 when K=3) from FAISS
- **Final output:** Top *K* results (e.g., 3) after filtering

5.2.1 Key Observations:

- If a relevant project is **not included in the initial FAISS candidate set**, it will not appear in the final results
- The fuzzy matching layer provides limited recovery and is primarily effective only when the query closely resembles **project IDs**, which is not typical for natural user queries

5.3 Chunking Strategy

Our analysis shows that the system does not implement a multi-chunking or sliding window strategy.

- Each project is represented as a **single consolidated "card"**, which is embedded as one vector

5.3.1 Implications:

- There is no risk of information fragmentation across multiple chunks
- However, retrieval performance is highly dependent on how comprehensively the single card representation captures all relevant project details

5.4 Context vs Metadata

The embedded content (project card) includes key information such as:

- Project name
- Location details
- Pricing (if available)
- Unit types and amenities

At the same time:

- Metadata (e.g., project_id) is stored separately and **not leveraged during semantic scoring**

5.4.1 Observation:

- While the embedded content is reasonably rich, retrieval is driven **entirely by embedding similarity**, without metadata-aware ranking or enrichment

5.5 Retrieval Approach & Duplicates

The current system uses:

- Dense vector search (FAISS)
- Supplemented by limited fuzzy matching

This does not constitute a full hybrid retrieval approach (i.e., no BM25 or sparse-dense score fusion).

- Duplicate results are **effectively handled through deduplication logic**, ensuring no repetition within the Top-K results

5.5.1 Structured Filters

The system currently supports the following structured filters:

- Region / location
- Project type
- Project status

However, key user-driven constraints are **not explicitly handled at the retrieval level**, including:

- Budget
- BHK / bedroom count
- Price ranges

These attributes are instead **implicitly handled through semantic matching**, which may not consistently enforce strict filtering requirements

5.5.2 Query Representation

User queries are processed through an LLM-based router that:

- Rewrites the query into a **clean, standalone format**
- Extracts a limited set of structured filters

The rewritten query is then used for embedding and retrieval.

5.5.3 Limitation:

- Only basic structured filters are captured
- More granular intent (e.g., budget constraints, unit preferences) is not translated into structured retrieval parameters

5.6 Root Cause Analysis

Based on our assessment, we summarize the system behavior as follows:

- **Reranker effectiveness:** Not applicable (no true reranking mechanism in place)
- **Chunking strategy:** Not a limiting factor (single-card representation is consistent)
- **Primary challenge: Retrieval quality**

5.6.1 Key contributing factors:

- Limited candidate pool (Top-2K before filtering)
- Absence of score-based ranking or fusion
- Lack of structured constraints (e.g., budget, BHK) in retrieval
- Strong reliance on embedding similarity against a fixed text template

6. Summary

From our evaluation, the current implementation functions as a **basic dense retrieval system with auxiliary fuzzy matching**, rather than a fully optimized RAG pipeline.

The **primary area for improvement lies in enhancing retrieval design**, particularly in terms of: